

SkillClaw: Let Skills Evolve Collectively with Agentic Evolver

Collective Skill Evolution for Multi-User LLM Agent Ecosystems

Source: arxiv.org/abs/2604.08377



Agent Skills Remain Static After Deployment — Users Rediscover Solutions Independently

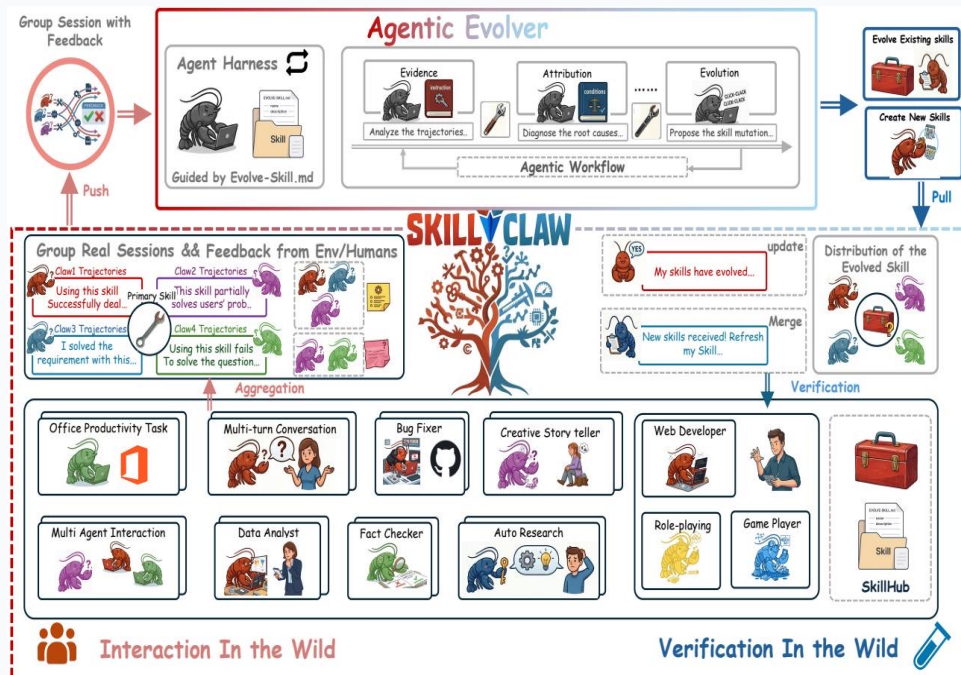
The Problem: Static Skills

- Manually installed skills never update from runtime experience.
- Failures and solutions are rediscovered repeatedly across users.
- No system-level knowledge accumulation.

Limitations of Existing Methods

- Memory-based methods remain tied to specific instances and are hard to generalize.
- Skill-based methods compress experience into structured instructions but treat the resulting library as static.

Closed-Loop Pipeline: From Multi-User Interaction to Shared Skill Evolution



Four Stages of Evolution

1. Multi-user Interaction

Agents produce structured session trajectories preserving action-feedback chains.

2. Session Collection

Trajectories are aggregated across users.

3. Agentic Evolver

Evidence → Attribution → Evolution.

4. Skill Synchronization

Verified skills merge to the shared repository.

Agentic Evolver: Evidence → Attribution → Evolution in Three Stages

Driven by an autonomous agent performing open-ended reasoning over interaction evidence, not predefined update rules.

1. Evidence

Analyze the trajectories to identify behavioral patterns. Aggregates data across the ecosystem.

2. Attribution

Diagnose root causes of failures and successes. Context-aware open-ended reasoning.

3. Evolution

Propose skill mutations — refining existing skills or creating new ones based on derived root causes.

Cross-User Trajectory Aggregation Exposes Skill Behavioral Boundaries

Compatible Claw-style Systems:

OpenClaw, CoPaw, IronClaw, PicoClaw, ZeroClaw, NanoClaw, NemoClaw

Complementary Signals

- Trajectories from different users exercising the same skill under diverse contexts produce complementary views of that skill's behavioral boundary.
- Reveals both conditions under which the skill works and those under which it breaks.

The Aggregation Advantage

- A single user rarely generates enough signal to separate a generalizable improvement from an idiosyncratic fix.

Case Study: Multimodal Creative Pipeline Skill — 6 Days of Evolution Attempts

Outcome: Only 1 of 6 candidates was accepted, illustrating the conservative verification process that prevents regressions.

Day	Candidate Skill	Validator	Outcome
1	validate-tmp-workspace-inputs	Accept	Upgrade to new best pool
2	multimodal-input-validation-and-setup	Reject	Keep current best pool
3	multimodal-creative-task-pipeline	Reject	Keep current best pool
4	multimodal-creative-task-pipeline (impr.)	Reject	Keep current best pool
5	pipeline + validate-required-input-files	Reject	Keep current best pool
6	multimodal-creative-task-pipeline (cand.)	Reject	Not admitted to next cycle

Case Study: Git Push Auth-Fallback Skill — Iterative Refinement Over 6 Days

Outcome: 3 of 6 days yielded accepted updates, showing successful iterative refinement before reaching a plateau.

Day	Candidate Skill	Validator	Outcome
1	git-push-with-auth-fallback (Safe fallback)	Accept	Add to Safety best pool
2	git-push-with-auth-fallback (Unified filenames)	Accept	Keep updated best pool
3	auth-fallback + git-clone-to-directory	Accept	Keep current best pool
4	(none) Same-pool retest	Accept	Continue same best pool
5	git-push-with-auth-fallback (Push hang)	Reject	Keep current best pool
6	git-push-with-auth-fallback (Identity config)	Reject	Not admitted to next cycle

Key Takeaways: Three Properties That Distinguish SkillClaw

1

Collective Evolution

Knowledge from individual interactions contributes to a shared, continuously improving skill ecosystem.

2

Fully Automatic

Skill evolution is driven by runtime interaction without manual curation or explicit user intervention.

3

Agentic Evolution Paradigm

Skill updates are produced through open-ended reasoning rather than predefined update rules.

Implication: Continuously improving agent systems avoid repetitive rediscovery and organically adapt to varying contexts.